

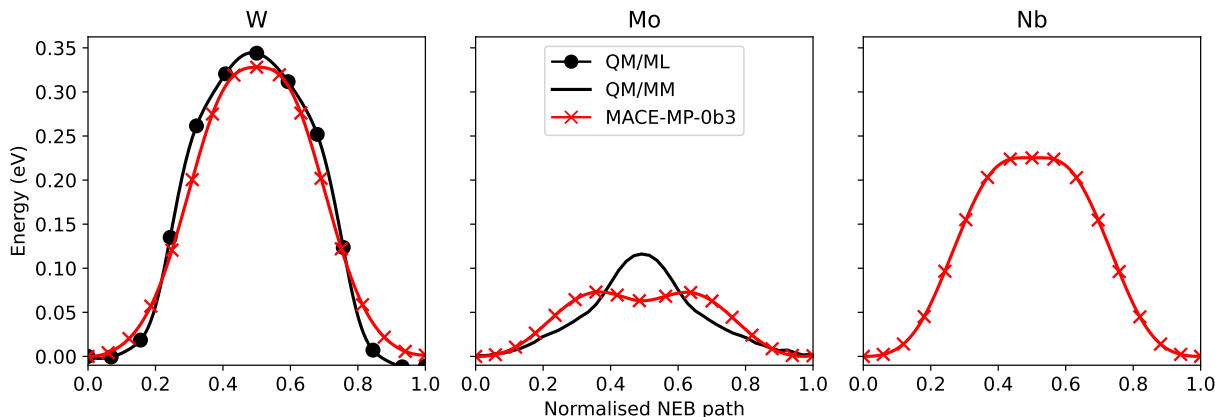


Figure S23: Edge $\langle 100 \rangle$ dislocation glide barriers for W, Mo and Nb. QM/ML data for W from Ref. (224), QM/MM data for Mo from Ref. (225)

Fine-tuning

Fine-tuning was performed on 142 W configurations containing isolated atom, deformed perfect BCC and FCC structures, vacancy in BCC and Molecular Dynamics snapshots for BCC. This is sufficient to correct softening of BCC and the over stability of FCC as seen in the updated elastic constants and equation of state plots in Fig. S18 and Fig. S19, and also improves the agreement of the lattice parameter and the vacancy and SIA formation energies as seen in Table S1 and Fig. S20. The improvement in the SIA formation energy demonstrates transferability as there are no interstitial configurations included in the fine-tuning dataset.

Similarity statement

The MP dataset includes 7 elemental tungsten, 7 elemental molybdenum, and 4 elemental niobium structures. They are all crystalline without any defects. Based on UMAP analysis, we find that the closest structures in the training set are mp-8641 for tungsten, mp-8637 for molybdenum and mp-8636 for niobium. We provide `W_input.json`, `Mo_input.json` and `Nb_input.json` to help visualize the interactive UMAP on chemiscope.org.

Performance summary

Energy-volume curves for BCC are well reproduced, while for FCC structures the MACE-MP-0b3 model shows a ca. 0.5 eV/atom shift in energy; this is corrected by fine-tuning. Stacking fault profile energies for Nb are well reproduced, while for W and Mo they are underestimated by a factor of around two with respect to DFT. Relaxed point defect structures are reasonable in all cases, with formation energies within 50% of reference DFT values (again, improved by fine tuning). Peierls energy barrier profiles for dislocation glide are qualitatively correct for the edge dislocations and underestimated for the screw dislocation together with incorrect screw dislocation core stability. There is a small spurious local minimum near the top of the barrier for the edge dislocation in Mo.

A.14 Alumina defects and bulk diffusion

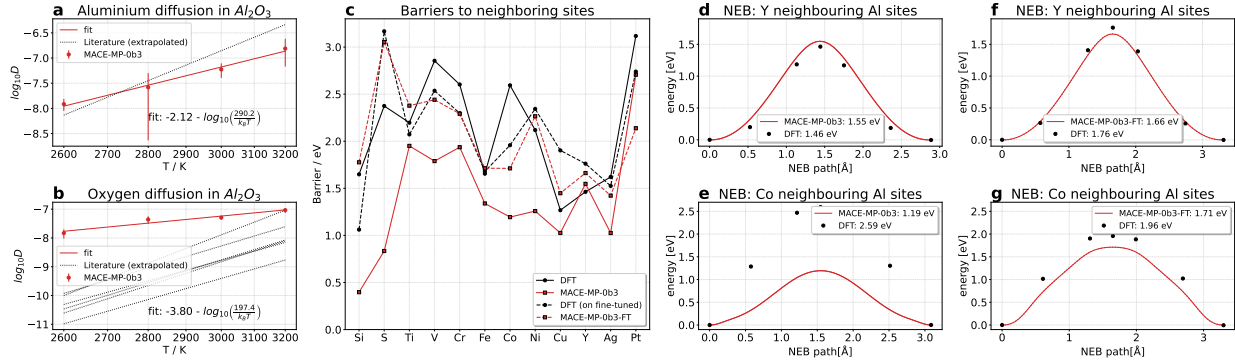


Figure S24: (a–b) Arrhenius plot of elemental diffusion in Al_2O_3 compared with experimental results from (227); (c) Comparison of MACE-MP-0b3 NEB barrier paths (red) and PBE single point evaluation of the MACE-MP-0b3 transition state (black) for elements in Al or O sites in Al_2O_3 moving to neighboring sites (connecting lines are guides to the eye), same protocol repeated with MACE-MP-0b3-FT as well shown with dashed lines; (d–e) lowest energy NEB path for Y (where MACE-MP-0b3 is accurate) and Co (where there is a substantial discrepancy), with single-point PBE evaluations on selected images of the obtained NEB path, indicated by black points; (f–g) Y and Co NEBs with the fine-tuned MACE-MP-0b3-FT model and the corresponding single point PBE evaluations.

A.14.1 Bulk diffusion

One Al and one O vacancy were introduced into a 270-atom alumina supercell, and over 2 ns (at 2.5 fs) MD was used at temperatures between 2600–3200 K to measure diffusivities of the two elements. Diffusivities agree within one order of magnitude for Al, and activation energies are underestimated for both compared to extrapolated experimental values (227). This demonstrates the long-timescale stability of the model, even at high temperatures and for long MD trajectories, but highlighting a shortcoming of the model for quantitative prediction of macroscopic observables.

A.14.2 Elemental defects

Elemental defects in Al_2O_3 were investigated by substituting Si, S, Ti, V, Cr, Fe, Co, Ni, Cu, Y, Ag, and Pt into lattice sites in a $2 \times 2 \times 1$ supercell (120 atom) and minimal energy paths to neighbouring sites were obtained using NEB (228, 229) starting from a linear interpolation. Paths were converged (max 100 steps, 0.3 eV/Å tolerance on projected forces) and the lowest energy one was tested with PBE single point evaluations using CASTEP (230). Comparing MACE-MP-0b3 and PBE on Fig. S24c–e there are large discrepancies, with a total force component RMSE of 0.35 eV/Å across 152 structures evaluated.

Finetuning

Fine-tuning was performed on 72 configurations, which included substitutions of each element and snapshots from paths found with MACE-MP-0b3. The barrier energies computed with the finetuned model are shown on Fig. S24c, showing improved agreement with the PBE reference in barrier energies and a lower 0.21 eV/Å force component RMSE on 149 PBE evaluations. Notably, the MACE-MP-0b3 model found a lower energy path for S, Ni, Cu, and Y than the finetuned model.

Similarity statement

There are 109 structures in the MP dataset containing exclusively Al & O, pure Al_2O_3 appears as mp-1143 (used to generate supercells). There is a total of 243 structures in the training set with Al & O and exactly

one of Si, S, Ti, V, Cr, Fe, Co, Ni, Cu, Y, Ag or Pt.

Performance summary

Activation energies for self-diffusivity are underestimated compared to experimental values (extrapolated from lower temperatures). Dopant atom migration minimum energy paths are all stable, and are sometimes accurate (e.g., Y) and sometimes only qualitative (e.g., Co) with respect to DFT single-point reevaluations. Fine-tuning substantially improves agreement with DFT.

A.15 Random structure search: Arsenic

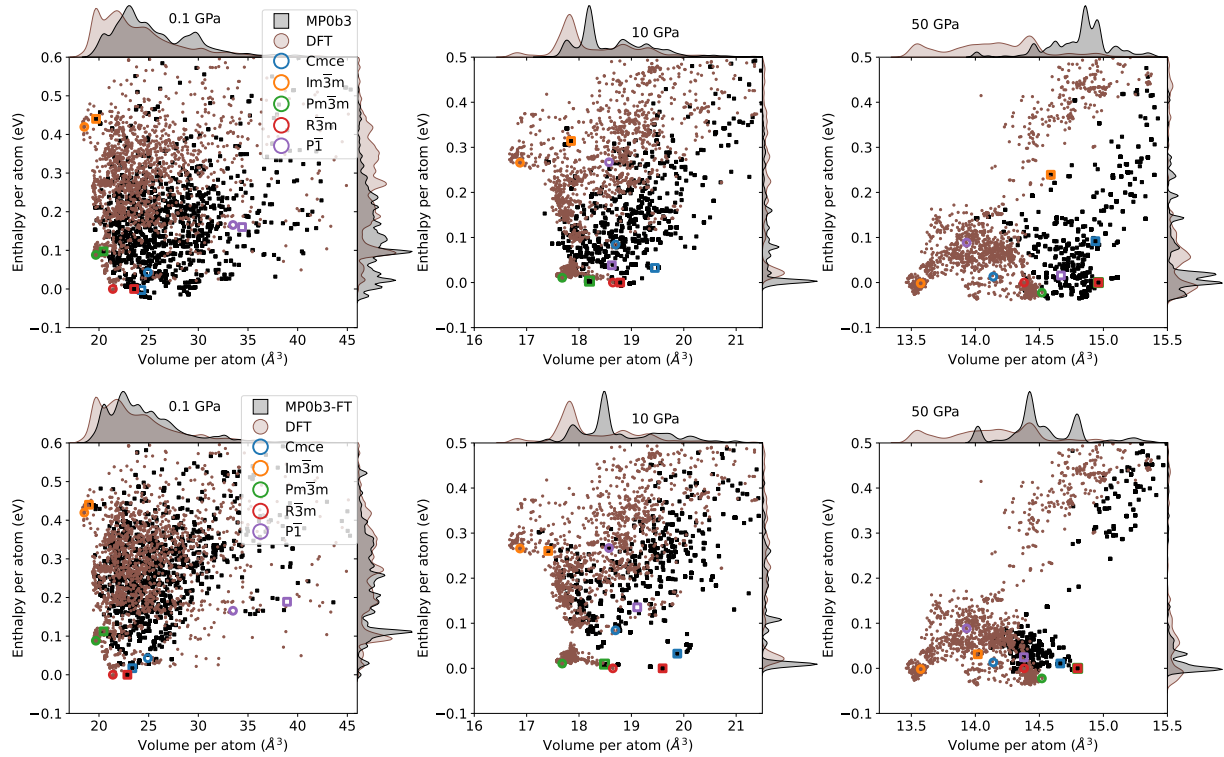


Figure S25: Densities of states of random structure search (RSS) minima for As at 0.1, 10 and 50 GPa obtained using DFT (brown, circles) and MACE-MP-0b3 (black, squares) before (top row) and after fine-tuning (bottom row). Known As structures are highlighted in various colors (see legend).

Ab initio Random Structure Searching (AIRSS) (231) is a simple, yet highly successful, approach for discovering new materials computationally. Multiple candidate structures are generated randomly, subject to physically motivated constraints, and then relaxed to local enthalpy minima using ab-initio methods such as DFT. There is great interest in accelerating structure prediction by using surrogate models (99, 232–234), such as ML potentials, to perform the initial structural relaxations. Here we test the suitability of MACE-MP-0 (without the D3 correction) for this task by searching for structures of Arsenic at 0.1, 10 and 50 GPa. The exceptional structural variety encountered during RSS probes the robustness of the model in an extremely extrapolative regime; there are only six As structures and no high-pressure data in the training set.

At each pressure, 2000 ($\times 100 n$) random structures were generated using $n = 2-6$ atoms per primitive unit cell, 2-4 randomly chosen symmetry operations, minimum distance constraints of $2 \text{ \AA}$ and a volume per atom of $15-40 \text{ \AA}^3$. The structures were then relaxed with MACE-MP-0 (ASE, force tolerance of $1 \times 10^{-3} \text{ eV/\AA}$) and CASTEP (230); PBE exchange-correlation functional (118), 400 eV cutoff energy, k-point spacing of $2\pi \times 0.05 \text{ \AA}^{-1}$ and Vanderbilt ultrasoft pseudopotentials (235) with a force tolerance of $0.05 \text{ eV/\AA}$ and stress tolerance of 0.1 GPa. With these settings 96% and 99% (at 0.1 and 10 GPa) of structural relaxations were successful with DFT and MACE-MP-0 respectively. The distributions of relaxed structures are depicted in Fig. S25 and the known structures listed in table S2 are highlighted with colored symbols. The energy and volume distributions are visually similar with the relative energy differences between the highlighted structures, particularly the low-energy ones, generally being small compared to the overall range. Fine-tuning improves the agreement further and reduces the systematic shift in volume distribution. Inspection of the structures at 0.1 GPa reveals that similar 3-fold coordinated 3D, layered, and 1D structures are found with both MACE-MP-0 and DFT. Furthermore, all known structures listed in table S2 were found using MACE-MP-0, including a simplified packing of the As_4 tetrahedra found in yellow As.

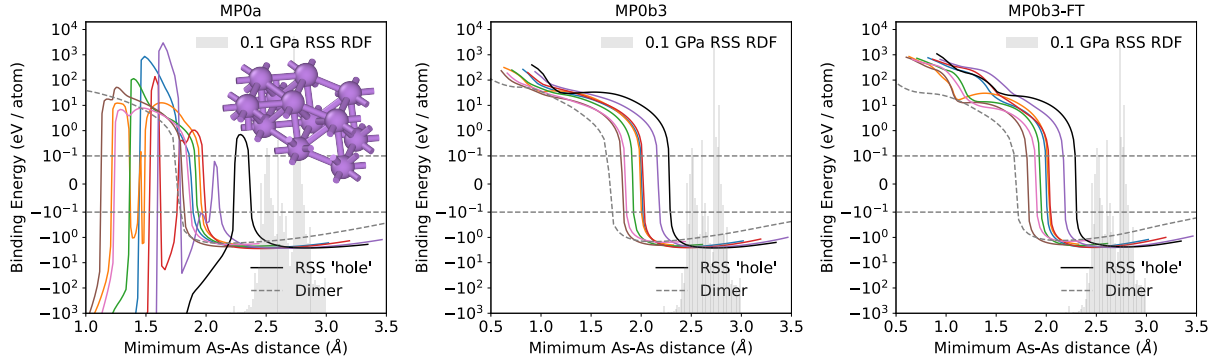


Figure S26: The binding energy of 7 (unrelaxed) randomly generated structures (colored lines), and one pathological structure found during RSS with MACE-MP-0a, the original version of our model, (black line and inset), are shown as the structures are compressed uniformly with fixed fractional atomic coordinates. The As–As dimer energy and the RDF for the 0.1 GPa As RSS structures are shown for comparison. Left panel corresponds to the original MACE-MP-0a model, the middle panel to MACE-MP-0b3, while the right panel to the fine-tuned MACE-MP-0b3-FT model.

The original MACE-MP-0a foundation model suffers from “holes” in the potential energy surface, where exceptionally dense, highly coordinated structures are predicted to be overly stable, as shown in Fig. S26. The repulsion seen in the As–As dimer curve indicates that these holes are caused by higher body-order terms in an extrapolative regime - holes occurs at shorter As–As distances than occur in the radial distribution function of the 0.1 GPa RSS results. These holes are typically not an issue during ambient pressure MD, due to the large energy barriers seen in Fig. S26. The updated model does not exhibit these holes due to the use of the ZBL repulsive pair potential and density normalisation (see Methods). Bar a systematic shift in the volume distribution, there is reasonable agreement with DFT even at 50 GPa.

Table S2: Summary of known Arsenic structures. The white P structure type is used as a proxy for yellow As as the structure is unknown (236).

Structure	Pressure	In training set?	Space Group	Z	Found with 0.1 GPa MACE-MP-0b3?
A7, grey As (237)	ambient	yes	$R\bar{3}m$	2	yes
black P (238)	ambient	yes	$Cmce$	4	yes
white P	ambient	no	$P\bar{1}$	24	As ₄ tetrahedra found
simple cubic (239)	27–57 GPa	yes	$Pm\bar{3}m$	1	yes
bcc (239)	≥ 110 GPa	no	$Im\bar{3}m$	1	yes

Fine-tuning

Fine-tuning was performed on a total of 200 configurations. The 5 known structures were relaxed at each pressure contributing 15 configurations to the fine-tuning. A further 185 configurations were selected from the relaxed structures from the original RSS using furthest-point sampling on averaged (across atomic sites) MACE descriptors. This procedure yielded a roughly even split between configurations at each pressure with 77, 51 and 72 configurations selected at 0.1, 10 and 50 GPa respectively.

Similarity statement

The MP dataset contains six pure As structures. Grey arsenic ($R\bar{3}m$) (237) and the orthorhombic allotrope ($Cmce$, isostructural with black phosphorus) (238) have been observed at ambient conditions whilst the simple cubic structure ($Pm\bar{3}m$) (239) is stable at moderate pressure between 27–57 GPa. The remaining